A Rayleigh Lower Bound for Normalized Inverse-GCD Matrices on Prime-Pair and Multi-Representation Vectors

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Disclaimer. Restricted Rayleigh bound on Goldbach test vectors; not a full-spectrum floor. No Navier–Stokes claim.

Abstract

Let $\widetilde{Q}_N(i,j) = 1/(\gcd(i,j)\sqrt{ij})$ for $1 \le i,j \le N$. For distinct primes p,q set $v = e_p - e_q$. We prove

$$R(v) = \frac{v^\top \widetilde{Q}_N v}{\|v\|_2^2} = \frac{1}{2}\Big(\frac{1}{p^2} + \frac{1}{q^2}\Big) - \frac{1}{\sqrt{pq}} \ > \ -\frac{1}{2}.$$

The same floor holds for multi-representation vectors $v_k = \sum_{p\in\mathcal{R}(k)}(e_p - e_{k-p})$ by a nonnegative cross-term identity. We do not claim a full-spectrum floor $\lambda_{\min}(\widetilde{Q}_N) > -1/2$ (false for moderate N).

1 Operator

$$\widetilde{Q}_N(p,p) = \frac{1}{p^2}, \qquad \widetilde{Q}_N(p,q) = \frac{1}{\sqrt{pq}} \quad (p \neq q \text{ primes}).$$

2 Single pair

Theorem 2.1. *For distinct primes p,q,*

$$R(e_p - e_q) = \frac{1}{2}\Big(\frac{1}{p^2} + \frac{1}{q^2}\Big) - \frac{1}{\sqrt{pq}} \ > \ -\frac{1}{2}.$$

Proof.

$$R + \frac{1}{2} = \frac{1}{2} + \frac{1}{2p^2} + \frac{1}{2q^2} - \frac{1}{\sqrt{pq}} > \frac{1}{2} - \frac{1}{\sqrt{pq}}.$$

Since $pq \ge 2\cdot 3 = 6$, one has $1/\sqrt{pq} \le 1/\sqrt{6} < 1/2$, hence $R + 1/2 > 0$. □

Remark 2.2. *The Rayleigh quotient may be negative (e.g. $(p,q) = (2,3)$ gives $R \approx -0.228$) but remains strictly above $-1/2$. Full-spectrum $\lambda_{\min}(\widetilde{Q}_N)$ falls below $-1/2$ already for small N; the present bound is restricted to Goldbach test vectors.*

3 Multi-representation

Lemma 3.1 (Cross-term factorization)**.** *For distinct primes $p \neq q$ and $r \neq s$,*

$$\widetilde{Q}(p,r) - \widetilde{Q}(p,s) - \widetilde{Q}(q,r) + \widetilde{Q}(q,s) = \big(p^{-1/2} - q^{-1/2}\big)\big(r^{-1/2} - s^{-1/2}\big).$$

If $p < q$ and $r < s$, the cross term is strictly positive.

Theorem 3.2 (Multi-representation Bridge*)**.** *Let k be even and $\mathcal{R}(k) = \{p \text{ prime} : 2 \leq p < k/2,\ k - p \text{ prime}\}$. Set $v_k = \sum_{p \in \mathcal{R}(k)} (e_p - e_{k-p}) \neq 0$. Then $R(v_k) > -1/2$.*

Proof. Unordered Goldbach pairs for fixed k are pairwise disjoint. Write $v_k = \sum_a v_a$ with $v_a = e_{p_a} - e_{q_a}$, $p_a < q_a$. Diagonal blocks give $R(v_a) > -1/2$ by the single-pair theorem; cross blocks are positive by the lemma. Hence $R(v_k) \geq \min_a R(v_a) > -1/2$. □

Full operator note: `04_q6_inverse_gcd.tex`.